# CS403P - Quiz No.8

Date: 19 December 2025

**Q.1 Suppose a relation named STUDENT contains attributes St_name, St_cgpa, cur_semester and st_roll. Which of the following query will be used to SELECT all those student records having cgpa greater than 3.0?**

**A. σcgpa>3.0 (STUDENT)**

B. Ωcgpa>3.0 (STUDENT)

C. €cgpa>3.0 (STUDENT)

D. σcgpa>3.0 (STUDENT)

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**Q.2 A relation named FACULTY contains the attributes FactId, FactName, FactDept and FactRank. Which of the query will be used to PROJECT the FactId and FactDept?**

A. σ FactId, FactDept (Faculty)

B. ϕ FactId, FactDept (Faculty)

C. Ω FactId, FactDept (Faculty)

**D. ∏ FactId, FactDept (Faculty)**

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**Q.3 In projection operation, any duplicate values are_____.**

A. Grouped

B. Copied

**C. Eliminated**

D. Selected

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**Q.4 If there are two relations X and Y then the difference of these two relations (i.e. X-Y) will be set of tuples that _____.**

A. Appear in both X and Y

**B. Appear in X but do not appear in Y**

C. Do not appear in both X and Y

D. Appear in Y but not in X

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**Q.5 Which of the following are unary operators?**

**A. SELECT and PROJECT**

B. SELECT and CARTESIAN

C. PROJECT and CARTESIAN

D. UNION and PROJECT

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**Q.6 Suppose a relation named EMPLOYEE contains attributes Emp_name, Emp_designation, Emp_address and Emp_salary. Which of the following query will be used to SELECT all those employees whose designation is Admin officer?**

**A. σ designation = 'Admin officer' (EMPLOYEE)**

B. α designation = 'Admin officer' (EMPLOYEE)

C.  designation = 'Admin officer' (EMPLOYEE)

D. ϕ designation = 'Admin officer'(EMPLOYEE)

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**Q.7 In intersection operation, which of the following is selected?**

**A. Tuples common in both relations**

B. Tuples of second relation

C. Tuples of first relation

D. Tuples of single relation only

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**Q.8 Suppose A → B, then which of the following is valid Functional dependency for augmentation?**

**A. AC → BC**
B. AB → C
C. A → C
D. AD → B

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**Q.9 2nd Normal Form is based on which of the following dependency?**

A. Reflexive dependency
B. Transitive dependency
**C. Partial dependency**
D. Full functional dependency

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**Q.10 If a relation CLASS (crId, stId, stName, fId, room, grade) has following Functional Dependencies: crId, stId → stName, fId, room, grade which of the following functional dependency is in above functional dependency?**

A. Reflexive dependency
B. Transitive dependency
**C. Squencial dependency**
D. Partial dependency

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**Q.11 _____ is a step by step process to reduce data redundancy and improve data integrity to build a more efficient and accurate database design.**

A. Classification
B. Integration
C. Modeling
**D. Normalization**

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**Q.12 In functional dependency, if one attribute is determinant of all other attributes of the relation then the relation has at least _____.**

**A. Super key**
B. Relational key
C. Random key
D. Secondary key

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**Q.13 Consider the following functional dependency: A → B In above functional dependency, attribute B is_____ on A.**

**A. Dependent**
B. Projected
C. Reflected
D. Terminated

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**Q.14 A relation is in second normal form if and only if _____ and all the non key attributes are fully functionally dependent on the key.**

A. It has multivalued attributes
B. It is not in first normal form
**C. It is in first normal form**
D. It has inconsistent data

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